# Space physics

Mark scheme and worked solutions for 30 questions: 15 multiple choice, 15 structured.

## 1 · Part 1 Core The Earth [1]

**Answer: C**

**Explanation**

The Earth spins on its axis once in about 24 hours, so the Sun appears to move across the sky. A is the old geocentric idea. B mixes up the orbit (about 365 days) with the rotation. D, the tilt, explains the seasons, not the daily motion.

## 2 · Part 1 Core The Earth [1]

**Answer: A**

**Explanation**

The Moon orbits the Earth in about one month, and the cycle of phases repeats once per orbit. B is the time from one quarter phase to the next. C is the Earth's rotation. D is the Earth's orbit of the Sun.

## 3 · Part 1 Core The Solar System [1]

**Answer: D**

**Explanation**

The order from the Sun is Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune. Venus, Mars, Jupiter, Uranus follows that order. A puts Saturn before Jupiter, B puts Earth before Venus, and C puts Neptune before Uranus.

## 4 · Part 1 Core The Solar System [1]

**Answer: B**

**Explanation**

Mars is one of the four inner planets, which are small and rocky. Neptune is one of the four outer planets, which are large and gaseous. A swaps the two planets. C and D pair the size with the wrong composition.

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## 5 · Part 1 Core The Solar System [1]

**Answer: A**

**Explanation**

The Sun's gravitational attraction on the Earth keeps it in orbit. The Moon's pull (B) is far too small to hold the Earth in an orbit around the Sun. Magnetism (C) is not what holds planets in orbit. Rotation (D) causes day and night and does not produce a force towards the Sun.

## 6 · Part 1 Supplement The Solar System [1]

**Answer: C**

**Explanation**

The Sun's gravitational field gets weaker as the distance from the Sun increases, and planets further from the Sun orbit more slowly. A and B wrongly say the field is stronger further away. D wrongly says outer planets move faster.

## 7 · Part 1 Supplement The Solar System [1]

**Answer: B**

**Explanation**

For an elliptical orbit the Sun is not at the centre. It sits inside the ellipse, closer to one end, so the comet's distance from the Sun changes around the orbit. The Sun is at the centre only when the orbit is almost circular, so A is wrong. The Sun is not on the comet's path (C) or outside the orbit (D).

## 8 · Part 1 Core The Sun as a star [1]

**Answer: D**

**Explanation**

The Sun gives out most of its energy as infrared, visible light and ultraviolet. A and C include gamma rays and X-rays, which the Sun gives out only in tiny amounts. B includes radio waves and microwaves, which also carry only a tiny fraction of its energy.

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## 9 · Part 1 Core Stars [1]

**Answer: A**

**Explanation**

A light-year is a distance: how far light travels through space in one year. B and C describe times. D describes a speed.

## 10 · Part 1 Supplement Stars [1]

**Answer: C**

**Explanation**

A very massive star swells into a red supergiant. It then explodes as a supernova and leaves behind a neutron star or a black hole. A is the sequence for a star like the Sun. B and D mix the two sequences together.

## 11 · Part 1 Core The Universe [1]

**Answer: B**

**Explanation**

Redshift is an increase in the observed wavelength of electromagnetic radiation from stars and galaxies that are moving away (receding). A describes a decrease in wavelength from objects moving closer, which is the opposite effect. C is wrong because a longer wavelength means a lower frequency, not a higher one. D mixes up redshift with a red giant changing colour.

## 12 · Part 1 Core The Universe [1]

**Answer: D**

**Explanation**

Light from almost all distant galaxies is redshifted, so they are moving away from us. This is evidence that the Universe is expanding, which supports the Big Bang Theory. A is the opposite of what redshift shows. B confuses redshift with the colour of cool stars. C is the opposite of expansion.

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## 13 · Part 1 Supplement The Universe [1]

**Answer: A**

### Explanation

The CMBR is microwave radiation of a specific frequency that is observed at all points in space around us. It does not come from the Sun (B) or from one direction (C). D is wrong because it is microwave radiation left over from soon after the Universe formed, not reflected infrared starlight.

## 14 · Part 1 Supplement The Universe [1]

**Answer: B**

### Worked solution

$v = H0 \times d = 2.2 \times 10^{-18} \times 4.0 \times 10^{24} = 8.8 \times 10^{6}$ m / s. Check: $8.8 \times 10^{6} / 4.0 \times 10^{24} = 2.2 \times 10^{-18}$ per second, which is the value of H0. The speed is about 3% of the speed of light, which is a sensible value for a far galaxy.

## 15 · Part 1 Supplement The Universe [1]

**Answer: C**

### Explanation

Supernovae are so bright that they can be seen in far galaxies, so the brightness of a supernova in a galaxy is used to determine the galaxy's distance. A radar echo (A) would be far too weak and would take millions of years to come back. The mass (B) does not give the distance. The CMBR (D) is observed at all points in space and does not depend on where any one galaxy is.

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## 16 · Part 1 Core The Earth [5]

**Mark scheme**

**(a)** **[1]**

- approximately 24 hours / one day (1)

**(b)** **[2]**

- the Earth rotates (spins) on its axis (1)
- the side of the Earth facing the Sun has day and the side facing away from the Sun has night; as the Earth rotates the city moves from one side to the other (1)

> **REJECT** the Earth orbits the Sun once a day

**(c)** **[2]**

- the Earth orbits the Sun once in approximately 365 days (one year) (1)
- the Earth's axis is tilted, so for part of the orbit the city's half of the Earth is tilted towards the Sun (summer: longer days, more direct sunlight) and for part it is tilted away from the Sun (winter) (1)

> **REJECT** summer happens because the Earth is closer to the Sun

**Worked solution**

(a) The Earth spins once on its axis in about 24 hours.

(b) At any moment, half of the Earth faces the Sun and has daylight, and the other half faces away and has night. As the Earth rotates, the city is carried from the sunlit side to the dark side and back again, so day and night repeat every 24 hours or so.

(c) The Earth goes once around the Sun in about 365 days. Because the Earth's axis is tilted, the city's half of the Earth leans towards the Sun for part of the orbit. The Sun is then higher in the sky and days are longer, so it is summer. Half a year later that half of the Earth leans away from the Sun and it is winter. One full orbit brings the same season back, so the pattern repeats every year. A common mistake is to say summer happens because the Earth is nearer the Sun. The tilt is what matters.

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## 17 · Part 1 Supplement The Earth [6]

**Mark scheme**

**(a)** **[1]**

- v = 2πr / T (1)

**ALLOW** orbital speed = 2π × average radius / orbital period

**(b)** **[3]**

- T = 3.55 × 24 × 3600 = 306 720 s ($3.07 \times 10^5$ s) (1)
- v = 2π × $6.7 \times 10^8$ / 306 720 (1)
- v = $1.4 \times 10^4$ m / s (1)

**ALLOW** 13 700 m / s or $1.37 \times 10^4$ m / s

**ALLOW** 14 km / s

**(c)** **[2]**

- T = 2πr / v = 2π × $5.4 \times 10^8$ / $1.5 \times 10^4$ = $2.26 \times 10^5$ s (1)
- T = $2.26 \times 10^5$ / 86 400 = 2.6 days (1)

**ECF** from an incorrect equation in (a) for the method mark only

**Worked solution**

(a) The average orbital speed is the distance around the orbit divided by the time for one orbit: v = 2πr / T.

(b) First change the period into seconds: 3.55 days × 24 h × 3600 s = 306 720 s. The distance around the orbit is 2π × $6.7 \times 10^8$ = $4.21 \times 10^9$ m. So v = $4.21 \times 10^9$ / 306 720 = 13 700 m / s, which is $1.4 \times 10^4$ m / s to 2 significant figures. Check: $1.37 \times 10^4$ m / s × 306 720 s = $4.21 \times 10^9$ m, which matches the distance around the orbit.

(c) Rearrange the equation: T = 2πr / v = 2π × $5.4 \times 10^8$ / $1.5 \times 10^4$ = $3.39 \times 10^9$ / $1.5 \times 10^4$ = 226 000 s. One day is 24 × 3600 = 86 400 s, so T = 226 000 / 86 400 = 2.6 days. The probe is closer to the planet than the moon and moves faster, so it is sensible that its period is shorter.

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## 18 · Part 1 Core The Solar System [5]

**Mark scheme**

**(a)** **[3]**

- time = distance / speed (in any form) (1)
- $t = 2.3 \times 10^{11} / 3.0 \times 10^{8} = 770$ s (767 s) (1)
- t = 13 min (1)

**ALLOW** 12.8 min

**ECF** correct conversion of an incorrect time in seconds into minutes

**(b)** **[2]**

- total distance = $2 \times 7.8 \times 10^{10}$ m = $1.56 \times 10^{11}$ m, OR one-way time = 260 s (1)
- time = 520 s (1)

**ALLOW** 8.7 min

**ALLOW** 1 mark for 260 s if no other mark is awarded

**Worked solution**

(a) Use speed = distance / time, so time = distance / speed. $t = 2.3 \times 10^{11}$ m / $3.0 \times 10^{8}$ m / s = 767 s. To change to minutes, divide by 60: 767 / 60 = 12.8 min, which is 13 min to 2 significant figures. Check: $3.0 \times 10^{8}$ m / s × 767 s = $2.3 \times 10^{11}$ m.

(b) The message has to travel from Mars to the Earth, and then the reply has to travel back from the Earth to Mars. The total distance is $2 \times 7.8 \times 10^{10} = 1.56 \times 10^{11}$ m. Time = $1.56 \times 10^{11} / 3.0 \times 10^{8} = 520$ s, which is about 8.7 minutes. A common mistake is to work out only the one-way time (260 s). Read the question carefully: the rover has to wait for the reply too.

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## 19 · Part 1 Core The Solar System [6]

**Mark scheme**

**(a)** **[2]**

- Venus is rocky and Saturn is gaseous (1)
- Venus is small and Saturn is large (1)

**(b)** *any four from:* **[4]**

- the Solar System formed from an interstellar cloud of gas and dust containing many elements (1)
- gravity pulled the material in the cloud together / the cloud collapsed due to gravity (1)
- the cloud was rotating and flattened into an accretion disc (1)
- most of the material collected at the centre and formed the Sun (1)
- material in the disc clumped together due to gravity to form the planets (1)
- near the Sun it was too hot for gases to collect, so only rocky material formed the inner planets, which are small; further out it was cooler, so large amounts of gas collected to form the large gaseous outer planets (1)

**ALLOW** gases were blown away from the region near the Sun

**Worked solution**

(a) Venus is small and rocky. Saturn is large and made mostly of gas. All four planets nearest the Sun are small and rocky, and all four furthest from the Sun are large and gaseous.

(b) The Solar System began as a huge interstellar cloud of gas and dust containing many different elements. Gravity pulled this material together and the cloud collapsed. The cloud was rotating, so as it collapsed it flattened into a spinning accretion disc. Most of the material fell to the centre and became the Sun. In the disc, gravity made small pieces of material clump together into bigger and bigger bodies, which became the planets. Close to the hot young Sun, gases could not collect, so only rocky material was left to form the inner planets, and they stayed small. Further out it was much colder, so the planets there gathered huge amounts of gas and became the large gaseous outer planets. To get full marks, mention gravity, the rotating cloud and the accretion disc, and then link them to the rocky inner planets and gaseous outer planets.

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## 20 · Part 1 Core The Solar System [5]

**Mark scheme**

**(a)** **[1]**

- the mass of the planet (1)

> **IGNORE** additional reference to the radius of the planet

**(b)** **[1]**

- greater at the surface of Jupiter (1)

**(c)** **[1]**

- it decreases (gets weaker) as the distance from the Earth increases (1)

> **REJECT** becomes zero as soon as the probe leaves the atmosphere

**(d)** **[2]**

- the Sun contains most of the mass of the Solar System (1)
- so the gravitational attraction of the Sun on each planet is the force that keeps the planet in orbit around it (1)

**Worked solution**

(a) The more mass a planet has, the stronger the gravitational field at its surface. So the field strength depends on the planet's mass.

(b) Jupiter has much more mass than the Earth, so the gravitational field strength at its surface is greater.

(c) A gravitational field gets weaker the further you are from the planet. As the probe moves away, the Earth's field at the probe keeps getting weaker. It never suddenly drops to zero.

(d) Almost all of the mass of the Solar System is in the Sun. This means the Sun's gravitational pull on every planet is much stronger than any other pull on it. That attraction towards the Sun is the force that keeps each planet moving in its orbit.

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## 21 · Part 1 Supplement The Solar System [7]

**Mark scheme**

**(a)** **[1]**

- the greater the orbital distance, the greater (longer) the orbital duration (1)

**REJECT** orbital duration is directly proportional to orbital distance

**(b)** **[2]**

- the average surface temperature decreases as the orbital distance increases (1)
- Venus (does not follow the trend: it is hotter than Mercury although it is further from the Sun) (1)

**(c)** **[3]**

- $r = 150 \times 10^6$ km $= 1.5 \times 10^{11}$ m (1)
- $T = 365 \times 24 \times 3600 = 3.15 \times 10^7$ s (1)
- $v = 2\pi r / T = 2\pi \times 1.5 \times 10^{11} / 3.15 \times 10^7 = 3.0 \times 10^4$ m / s (1)

**ALLOW** $2.99 \times 10^4$ m / s; ALLOW: 365.25 days

**ECF** final mark for correct use of $v = 2\pi r / T$ with an incorrectly converted r or T

**(d)** **[1]**

- Jupiter has a much greater mass than the Earth (1)

**IGNORE** references to Jupiter being gaseous

**Worked solution**

(a) Read down the table. As the distance from the Sun goes up, the time for one orbit also goes up, and it goes up by more than the distance does. Saturn is about 10 times as far from the Sun as the Earth, but it takes about 30 years to orbit. So the relationship is not direct proportion.

(b) Apart from one planet, the temperature falls steadily as you move away from the Sun: 167 °C, then 15 °C, −65 °C, −110 °C and −140 °C. Venus breaks the pattern. It is further from the Sun than Mercury but it is hotter, at 464 °C.

(c) Change the units first. $150 \times 10^6$ km $= 150 \times 10^9$ m $= 1.5 \times 10^{11}$ m. The Earth's orbital duration is 1.00 year, which is about 365 days: $365 \times 24 \times 3600 = 3.15 \times 10^7$ s. Then $v = 2\pi r / T = (2 \times \pi \times 1.5 \times 10^{11}) / 3.15 \times 10^7 = 9.42 \times 10^{11} / 3.15 \times 10^7 = 2.99 \times 10^4$ m / s, about $3.0 \times 10^4$ m / s (30 km / s). Check: $3.0 \times 10^4$ m / s for $3.15 \times 10^7$ s covers $9.4 \times 10^{11}$ m, which is the distance around the orbit.

(d) The gravitational field strength at the surface of a planet depends on the planet's mass. Jupiter has far more mass than the Earth, so its surface field strength is much greater.

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## 22 · Part 1 Supplement The Solar System [5]

**Mark scheme**

**(a)** **[1]**

- point P (1)

**(b)** **[3]**

- as the comet moves from Q to P it gets closer to the Sun, so its gravitational potential energy decreases (1)
- the total energy of the comet is conserved / stays the same (1)
- so its kinetic energy increases and the comet speeds up (1)

> **ALLOW** gravitational potential energy is transferred to kinetic energy for the second and third marks if both ideas are clearly present

**(c)** **[1]**

- its distance from the Sun stays (almost) the same, so its gravitational potential energy (and so its kinetic energy) stays (almost) constant (1)

**Worked solution**

(a) An object in an elliptical orbit moves fastest when it is closest to the Sun, which is at P.

(b) As the comet falls inwards from Q towards P, it gets closer to the Sun, so its gravitational potential energy decreases. Energy cannot be created or destroyed, so the total energy of the comet stays the same. The lost gravitational potential energy becomes kinetic energy, so the comet speeds up. On the way back out from P to Q the opposite happens and it slows down.

(c) In an almost circular orbit the planet stays at almost the same distance from the Sun all the time. Its gravitational potential energy hardly changes, so its kinetic energy hardly changes, and its speed stays almost constant.

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## 23 · Part 1 Supplement The Sun as a star [5]

**Mark scheme**

**(a)** **[1]**

- hydrogen and helium (both needed) (1)

**(b)** **[2]**

- (nuclear) fusion (1)
- hydrogen (nuclei) join together to form helium (releasing energy) (1)

**REJECT** fission for the first mark

**(c)** **[2]**

- there is an inward force due to gravitational attraction (1)
- which is balanced by an outward force due to the high temperature in the centre of the star (1)

**ALLOW** outward force due to (thermal) pressure from the hot centre

**Worked solution**

(a) The Sun is made mostly of hydrogen and helium.

(b) In a stable star the energy comes from nuclear fusion. Hydrogen nuclei join together to make helium nuclei, and energy is released. Do not confuse this with fission, which is the splitting of large nuclei.

(c) Gravity pulls all the material of the Sun inwards towards its centre. The very high temperature at the centre produces an outward force. In a stable star these two forces are balanced, so the star does not collapse and does not expand.

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## 24 · Part 1 Supplement Stars [7]

**Mark scheme**

**(a)** **[2]**

- the cloud (of gas and dust containing hydrogen) collapses due to its own (internal) gravitational attraction (1)
- the temperature of the collapsing cloud increases (1)

**(b)** **[2]**

- star A expands to form a red giant (1)
- which then forms a planetary nebula with a white dwarf star at its centre (1)

**(c)** **[2]**

- star B expands to form a red supergiant (1)
- which explodes as a supernova, leaving a neutron star or a black hole (at the centre of a nebula) (1)

> **ALLOW** forms a nebula containing hydrogen and new heavier elements for the first mark if red supergiant is not mentioned

**(d)** **[1]**

- $640 \times 9.5 \times 10^{15} = 6.1 \times 10^{18}$ m (1)

> **ALLOW** $6.08 \times 10^{18}$ m

**Worked solution**

(a) An interstellar cloud of gas and dust, which contains hydrogen, starts to collapse because of the gravitational attraction between its parts. As it collapses it heats up. This collapsing, heating cloud is a protostar.

(b) When most of the hydrogen in its centre has turned into helium, a star like the Sun swells up into a red giant. Later its outer layers drift away as a planetary nebula, and a small, hot white dwarf is left at the centre.

(c) A much more massive star swells into a red supergiant instead. This explodes as a supernova. The explosion makes a nebula containing hydrogen and new heavier elements, and leaves a neutron star or a black hole at the centre.

(d) One light-year is $9.5 \times 10^{15}$ m, so 640 light-years = $640 \times 9.5 \times 10^{15} = 6.08 \times 10^{18}$ m, which is $6.1 \times 10^{18}$ m to 2 significant figures.

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## 25 · Part 1 Core The Universe [4]

**Mark scheme**

**(a)** **[1]**

- the Milky Way (1)

**(b)** **[1]**

- (approximately) 100 000 light-years (1)

**(c)** **[1]**

- many billions (1)

**ALLOW** billions / more than a billion

**(d)** **[1]**

- the other stars are much further away from the Earth than the Sun is (1)

**Worked solution**

(a) The Sun is one of the stars in the galaxy called the Milky Way.

(b) The Milky Way is about 100 000 light-years across. This means light takes about 100 000 years to cross it.

(c) The Milky Way is just one of many billions of galaxies in the Universe.

(d) The Sun is our nearest star. Every other star in the Milky Way is much, much further from the Earth than the Sun is.

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## 26 · Part 1 Core The Universe [5]

**Mark scheme**

**(a)** **[2]**

- an increase in the observed wavelength (of electromagnetic radiation / light) (1)
- from stars or galaxies that are moving away from the Earth (receding) (1)

**(b)** **[1]**

- $6.89 \times 10^{-7} - 6.56 \times 10^{-7} = 3.3 \times 10^{-8}$ m (1)

**ALLOW** $0.33 \times 10^{-7}$ m

**(c)** **[2]**

- (redshift shows that) distant galaxies are moving away from the Earth (1)
- so the Universe is expanding, which suggests it started from a much smaller region / it began with a Big Bang (1)

**Worked solution**

(a) Redshift is an increase in the wavelength we observe for light (or other electromagnetic radiation) from a star or galaxy that is moving away from us.

(b) Increase in wavelength = observed wavelength − laboratory wavelength = $6.89 \times 10^{-7} - 6.56 \times 10^{-7} = 0.33 \times 10^{-7}$ m $= 3.3 \times 10^{-8}$ m. Check: $6.56 \times 10^{-7} + 0.33 \times 10^{-7} = 6.89 \times 10^{-7}$ m. The wavelength has got longer, which is what redshift means.

(c) Because the light from almost all distant galaxies is redshifted, those galaxies are moving away from us. This means the Universe is expanding. If it is expanding now, it must have been much smaller in the past, which is what the Big Bang Theory says.

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## 27 · Part 1 Supplement The Universe [4]

**Mark scheme**

**(a)** **[1]**

- it is observed at all points in space around us / from every direction (1)

**(b)** **[3]**

- the CMBR was produced shortly after the Universe was formed (1)
- as the Universe expanded, the radiation was stretched / its wavelength increased (1)
- (so it has been expanded) into the microwave region of the electromagnetic spectrum (1)

**REJECT** it is emitted by stars today

**Worked solution**

(a) The CMBR arrives from every direction and is observed at all points in space around us. Radiation from a single star or galaxy would come from one direction only.

(b) The CMBR was produced shortly after the Universe was formed. Since then the Universe has been expanding, and the expansion has stretched the radiation so that its wavelength has increased. It has now been stretched into the microwave region of the electromagnetic spectrum, which is why we detect it as microwaves.

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## 28 · Part 1 Supplement The Universe [7]

**Mark scheme**

**(a)** **[1]**

- from the change in wavelength of the galaxy's starlight due to redshift (1)

**(b)** **[1]**

- the ratio of the speed at which a galaxy is moving away from the Earth to its distance from the Earth (1)

**ALLOW** $H0 = v / d$ with v and d defined

**(c)** **[2]**

- $H0 = v / d = 6.9 \times 10^{6} / 3.0 \times 10^{24}$ (1)
- $H0 = 2.3 \times 10^{-18}\ s^{-1}$ / per second (1)

**(d)** **[3]**

- $d = v / H0 = 4.4 \times 10^{6} / 2.2 \times 10^{-18}$ (1)
- $d = 2.0 \times 10^{24}$ m (1)
- $d = 2.0 \times 10^{24} / 9.5 \times 10^{15} = 2.1 \times 10^{8}$ light-years (1)

**ECF** third mark for correct conversion of an incorrect distance in metres

**ALLOW** $1.9 \times 10^{24}$ m and $2.0 \times 10^{8}$ light-years if the value of H0 from (c) is used (max 2)

**Worked solution**

(a) Light from a galaxy that is moving away is redshifted. The bigger the change in wavelength, the faster the galaxy is moving away, so the change in wavelength gives the speed.

(b) The Hubble constant is the ratio of the speed at which a galaxy is moving away from the Earth to its distance from the Earth: $H0 = v / d$.

(c) $H0 = v / d = 6.9 \times 10^{6} / 3.0 \times 10^{24} = 2.3 \times 10^{-18}$ per second. The unit is (m / s) / m = 1 / s, which is $s^{-1}$. This is close to the current estimate of $2.2 \times 10^{-18}$ per second.

(d) Rearrange $H0 = v / d$ to give $d = v / H0$. $d = 4.4 \times 10^{6} / 2.2 \times 10^{-18} = 2.0 \times 10^{24}$ m. One light-year is $9.5 \times 10^{15}$ m, so $d = 2.0 \times 10^{24} / 9.5 \times 10^{15} = 2.1 \times 10^{8}$ light-years. Check: $2.2 \times 10^{-18} \times 2.0 \times 10^{24} = 4.4 \times 10^{6}$ m / s, which is the speed given.

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## 29 · Part 1 Supplement The Universe [6]

**Mark scheme**

**(a)** **[1]**

- $2.2 \times 10^{-18}$ per second / $s^{-1}$ (value and unit needed) (1)

**(b)** **[2]**

- age = d / v = 1 / H0 = 1 / $2.2 \times 10^{-18}$ (1)
- = $4.5 \times 10^{17}$ s (1)

**ECF** from (a)

**(c)** **[2]**

- one year = $365 \times 24 \times 3600 = 3.15 \times 10^{7}$ s (1)
- $4.5 \times 10^{17} / 3.15 \times 10^{7} = 1.4 \times 10^{10}$ years (1)

**ALLOW** 365.25 days; ECF from (b)

**(d)** **[1]**

- d / v is the time for a galaxy moving at speed v to travel distance d, so (running the expansion backwards) all galaxies / all the matter would have been at the same point at that time (1)

**Worked solution**

(a) The current estimate of the Hubble constant is $2.2 \times 10^{-18}$ per second.

(b) For any galaxy, d / v is the time it has taken to reach its distance d travelling at speed v. Since H0 = v / d, this time is 1 / H0 = 1 / $2.2 \times 10^{-18}$ = $4.5 \times 10^{17}$ s.

(c) One year is $365 \times 24 \times 3600 = 3.15 \times 10^{7}$ s. Age = $4.5 \times 10^{17} / 3.15 \times 10^{7} = 1.4 \times 10^{10}$ years, which is about 14 billion years. Check: $1.4 \times 10^{10} \times 3.15 \times 10^{7} = 4.4 \times 10^{17}$ s, which agrees with (b) to within rounding.

(d) Because H0 is the same for every galaxy, d / v gives the same time for every galaxy. Run the expansion backwards for this time and every galaxy arrives back at the same place. So at that time all the matter in the Universe was together at a single point.

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## 30 · Part 1 Supplement The Universe [6]

**Mark scheme**

**(a)** **[1]**

- its brightness (1)

**(b)** **[1]**

- the change in wavelength (redshift) of the light (1)

**(c)** **[3]**

- time = distance / speed of light, with speed of light = $3.0 \times 10^8$ m / s (1)
- $t = 1.2 \times 10^{25} / 3.0 \times 10^8 = 4.0 \times 10^{16}$ s (1)
- $t = 4.0 \times 10^{16} / 3.15 \times 10^7 = 1.3 \times 10^9$ years (1)

**ALLOW** distance in light-years = $1.2 \times 10^{25} / 9.5 \times 10^{15} = 1.3 \times 10^9$ light-years, so $1.3 \times 10^9$ years ago (3 marks)

**ECF** final mark for correct conversion of an incorrect time in seconds

**(d)** **[1]**

- supernovae are extremely bright, so they can be seen at very great distances where ordinary stars are too faint to be seen (separately) (1)

**Worked solution**

(a) Astronomers measure how bright the supernova appears. Comparing this with how bright supernovae really are tells them how far away the galaxy is.

(b) The speed is found from the change in wavelength (the redshift) of the galaxy's light.

(c) We see the supernova when its light reaches us, so the explosion happened as long ago as the light took to travel. Time = distance / speed = $1.2 \times 10^{25} / 3.0 \times 10^8 = 4.0 \times 10^{16}$ s. One year is $3.15 \times 10^7$ s, so the time is $4.0 \times 10^{16} / 3.15 \times 10^7 = 1.3 \times 10^9$ years. Another way: the distance in light-years is $1.2 \times 10^{25} / 9.5 \times 10^{15} = 1.3 \times 10^9$ light-years, and light takes one year to travel each light-year, so the answer is again $1.3 \times 10^9$ years.

(d) A single ordinary star in a far galaxy is much too faint to see on its own. A supernova is so bright that it can be seen even in very distant galaxies, so it can be used to measure their distances.

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